Plastic Pollution and Recycling

Introduction to Plastic Pollution

Plastic pollution is the accumulation of plastic waste in the environment, on land and in rivers, oceans and even the air, to the extent that it harms wildlife, ecosystems and human health. Plastics are synthetic or semi-synthetic polymers made mostly from petroleum. They are cheap, light, strong and durable, which makes them useful, but that same durability makes plastic waste last for a very long time. The first fully synthetic plastic, Bakelite, was invented by Leo Baekeland in 1907. Global plastic production has grown from about 2 million tonnes in 1950 to over 400 million tonnes a year today. Less than 10 percent of all plastic ever made has been recycled, and millions of tonnes enter the oceans every year. Plastic pollution has been called one of the major environmental problems of our time.

How Long Does Plastic Take to Decompose?

Plastic does not biodegrade easily, which means microbes cannot break it down quickly like food or paper. Instead it slowly fragments into smaller pieces by sunlight and weathering, a process called photodegradation. The decomposition time depends on the type and environment. Estimates suggest a plastic bag may take 10 to 20 years or more to break up, a plastic bottle about 450 years, and fishing line around 600 years. In general, plastics take from about 20 years to hundreds of years, and some may never fully decompose, only becoming smaller. These are estimates, since modern plastics have existed for less than a century. Plastic in landfills is buried and degrades even more slowly. Burning plastic in open air releases toxic gases such as dioxins.

What Are Microplastics?

Microplastics are tiny plastic particles less than 5 millimetres in size, and nanoplastics are even smaller, below 1 micrometre. Primary microplastics are made small on purpose, such as microbeads in some face scrubs and toothpastes, and plastic pellets used in industry. Secondary microplastics form when larger plastic items such as bottles, bags, fishing nets and tyres break down. Synthetic clothes like polyester and nylon shed microfibres during washing, and vehicle tyres release particles as they wear. Microplastics have been found in the deepest ocean trenches, in Arctic ice, in drinking water, sea salt, fish and seafood, in the air, and even in human blood, lungs and placentas. Scientists are still researching their health effects, which may include inflammation and the carrying of toxic chemicals. They harm marine animals that mistake them for food.

Plastic Recycling Codes

Most plastic products carry a number from 1 to 7 inside a triangle of arrows, called the resin identification code, which was introduced in 1988 by the Society of the Plastics Industry in the United States. The code tells the type of plastic, and not necessarily that the item is recyclable. Code 1 is PET or PETE (polyethylene terephthalate), used in water and soft drink bottles, and is widely recycled. Code 2 is HDPE (high-density polyethylene), used in milk cans, detergent bottles and pipes. Code 3 is PVC (polyvinyl chloride), used in pipes and cables, which is hard to recycle and may release harmful chemicals. Code 4 is LDPE (low-density polyethylene), used in carry bags and films. Code 5 is PP (polypropylene), used in food containers, bottle caps and furniture. Code 6 is PS (polystyrene), used in foam cups, trays and packing, which is rarely recycled. Code 7 is "other", covering polycarbonate, mixed plastics and bioplastics such as PLA, and is the hardest to recycle.

Recycling Methods

Mechanical recycling sorts, cleans, shreds and melts plastic into pellets for new products. Chemical recycling breaks plastic into its basic chemicals. Energy recovery burns waste for energy. Plastic waste is also used in road construction: Professor R. Vasudevan of Thiagarajar College of Engineering in Madurai, Tamil Nadu, developed a method of using shredded plastic waste mixed with bitumen to build roads, and he is called the Plastic Man of India. Many thousands of kilometres of such roads have been built in India. Reduce, Reuse, Recycle is the guiding principle, and reducing use is the most effective step.

Plastic Pollution in India

India generates roughly 3.5 to 4 million tonnes of plastic waste each year, according to the Central Pollution Control Board. Plastic clogs drains and worsens floods, harms cattle and marine life, and ends up in rivers like the Ganga and the Adyar. India hosted World Environment Day in 2018 on the theme "Beat Plastic Pollution". The Plastic Waste Management Rules, 2016, with later amendments, introduced Extended Producer Responsibility (EPR), under which manufacturers and brand owners are responsible for collecting and recycling their packaging. A ban on identified single-use plastic items, such as plastic cutlery, straws, earbuds and thin carry bags, came into force across India on 1 July 2022. Tamil Nadu banned 14 types of single-use plastics from 1 January 2019, and launched the "Meendum Manjappai" campaign in December 2021 to encourage cloth bags.

Global Action

Over 170 countries agreed in 2022 to develop a legally binding global treaty on plastic pollution, and negotiations by the United Nations have continued in several rounds, including in Busan in 2024 and Geneva in 2025, without final agreement.

What Individuals Can Do

Carry a cloth bag and a steel bottle, refuse plastic straws and cutlery, buy products with less packaging, segregate dry and wet waste at home, avoid burning plastic, and join beach and river clean-ups.

Challenges and Future

Challenges include low recycling rates, mixed and contaminated waste and the lack of cheap alternatives. Solutions under research include biodegradable plastics, plastic-eating enzymes, better waste collection and a circular economy where plastic is kept in use.

Quick Facts

- Plastic can take from about 20 to 600 years or more to break down; a plastic bottle about 450 years.
- Microplastics are plastic particles smaller than 5 mm.
- Recycling codes 1 to 7 are PET, HDPE, PVC, LDPE, PP, PS and Other.
- India banned identified single-use plastics from 1 July 2022.
- Prof. R. Vasudevan of Madurai developed plastic roads and is known as the Plastic Man of India.